GENERATIVE AI · CODING TRACK

Generative AI Developer

Build GenAI apps with Python, LLM APIs and RAG, and deploy them.

TRACK	DURATION	FORMAT	COURSE FEE
Coding	3 to 6 months	Live online, weekdays, small batch	₹15,000 (full fee)

WHO IT IS FOR
Engineering students, developers, data analysts and IT professionals moving into AI.

BEFORE YOU JOIN
Basic computer skills. Python is taught from the start; any prior coding helps.

What you will be able to do

- ✔ Write Python for AI applications
- ✔ Call OpenAI, Gemini and Claude APIs and run open-source LLMs
- ✔ Design prompts and structured outputs for apps
- ✔ Build RAG chatbots over your own documents
- ✔ Evaluate and improve LLM answers
- ✔ Deploy a GenAI app as an API on Google Cloud

Course path: 12 modules

1 Module 1: Python for AI

- Python basics, functions and OOP
- JSON handling and files
- Calling REST APIs
- Virtual environments and packages

Hands-on: A Python CLI chatbot skeleton

2 Module 2: How LLMs work

- Tokens, embeddings and context windows
- Temperature and sampling
- Popular models: GPT, Gemini, Claude, Llama
- Choosing a model: cost, speed, quality

Hands-on: Compare 3 models on the same tasks

3 Module 3: LLM APIs

- OpenAI, Gemini and Claude APIs
- System prompts and chat history
- Streaming responses
- API keys, limits and cost control

Hands-on: A chatbot that calls an LLM API

4 Module 4: Prompt engineering for developers

- Zero-, one- and few-shot prompting
- Chain of thought and prompt chaining
- Structured output and JSON schemas
- Prompt templates in code

Hands-on: Extract structured data from text into JSON

5 Module 5: Open-source LLMs

- Running models locally with Ollama
- Hugging Face models and pipelines
- When to use open vs hosted models
- Hardware basics

Hands-on: Run a local LLM and build a private chatbot

6 Module 6: Embeddings and vector databases

- What embeddings are
- Similarity search
- Chroma and Pinecone
- Metadata and filtering

Hands-on: Semantic search over a set of documents

7 Module 7: RAG: chat with your documents

- Why RAG
- Loading and chunking documents
- Retrieval and answer generation
- Citations and source tracking

Hands-on: A PDF / document chatbot

8 Module 8: LangChain for GenAI apps

- Prompts, chains and output parsers
- Retrievers and memory
- Building multi-step pipelines
- Debugging and tracing

Hands-on: Resume analyzer with LangChain

9 Module 9: Fine-tuning basics

- When to fine-tune vs prompt vs RAG
- Preparing training data
- Fine-tuning a small model
- Comparing results

Hands-on: Fine-tune a small model for a classification task

10 Module 10: Evaluation and safety

- Testing LLM outputs
- Hallucination checks
- Prompt injection and input validation
- Content safety and guardrails

Hands-on: An evaluation script for your chatbot

11 Module 11: Deployment

- Building an API with FastAPI
- Docker basics
- Deploying on Google Cloud
- Logging and monitoring costs

Hands-on: Deploy your GenAI app as an API

12 Module 12: Capstone project

- Choose a real problem
- Design, build and test
- Deploy and document
- Present your project

Hands-on: Capstone: a deployed GenAI app for your portfolio and GitHub

Tools you will use

Python · VS Code · Google Colab · OpenAI API · Gemini API · Claude API · Ollama · Hugging Face · LangChain · Chroma · Pinecone · FastAPI · Docker · Google Cloud · GitHub

How every module works

1	2	3	4	5	6
Learn the concept	See a live demo	Practise together	Build a project	Get feedback	Add to portfolio

What is included

Live interactive classes · Small batch · Hands-on project in every module · Course materials · Feedback on your work · Resume and interview guidance · Certificate of Completion · Free Peaceful Living course

Reserve your seat

Message me for the next batch dates and class timings.

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Trainer: Bubna Kallidukkil

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